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RESULTS

Results

01 · ONE-WAY ANOVA + POST-HOC

do 3+ groups differ, and which pairs?

Table 1. Descriptives by group

Group	N	Mean	Std. Dev.
Blended	80	65.49	6.85
Flipped	80	70.75	6.58
Lecture	80	60.46	7.57

Table 2. ANOVA

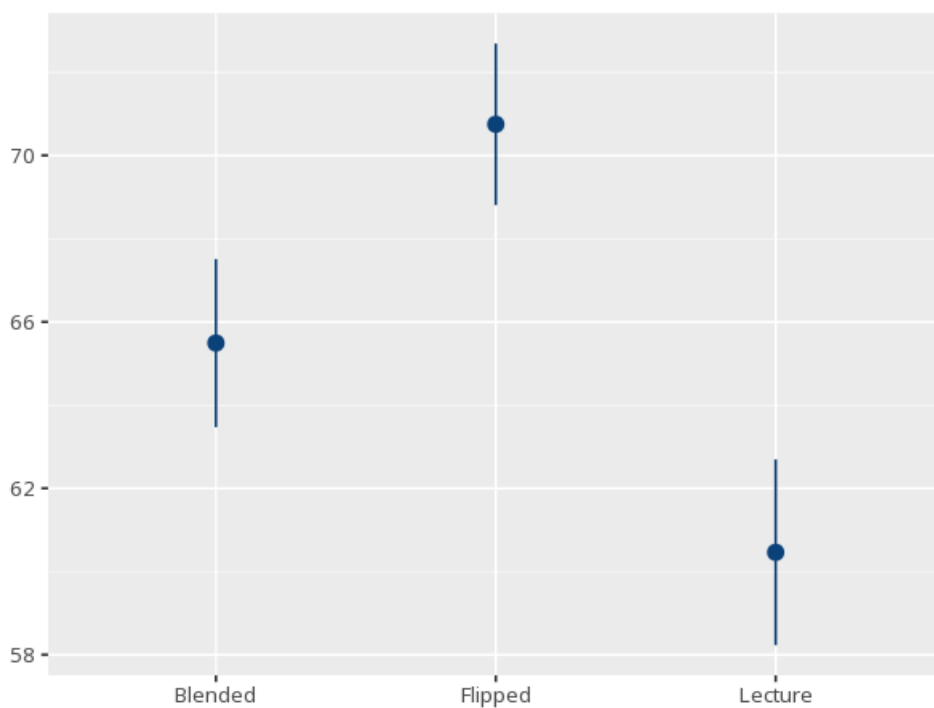
Source	SS	df	MS	F	p	η^2 [99% CI]
Between	4234.40	2	2117.20	43.07	<.001	0.27 [0.16, 1.00]
Within	11650.16	237	49.16			

Table 3. Post-hoc comparisons

Pair	M_{diff}	SE	P_{adj}	99% CI
Blended - Flipped	-5.26	1.11	<.001	[-8.52, -2.00]
Blended - Lecture	5.03	1.11	<.001	[1.77, 8.29]
Flipped - Lecture	10.29	1.11	<.001	[7.03, 13.55]

assumption checks: Levene's (equal variances) & normality of residuals. (Levene $F=1.44$, $p=.239$ · Shapiro $W=0.99$, $p=.017$) - equal variances look reasonable - residual normality looks doubtful; consider a nonparametric alternative (e.g. Kruskal-Wallis) or interpreting with caution

Figure. Group means



Understanding the numbers

F. The ratio of between-group to within-group variance - larger means the groups differ more than chance predicts. Here, $F(2, 237) = 43.07$.

η^2 . The proportion of total variance explained by group membership ($\sim .01$ small, $.06$ medium, $.14$ large). Here, $\eta^2 = .27$ $[.16, 1.00]$.

p. The probability of an F this large (or larger) if all group means were truly equal. Here, $p < .001$.

Source. Which portion of total variance a row summarizes: between-groups (explained by the factor) or within-groups (residual/error). Here, the Between row captures variance explained by the grouping factor; the Within row captures unexplained (residual) variance.

SS. The sum of squares - the raw amount of variability attributed to a source. Here, $SS = 4234.40$ (Between) and 11650.16 (Within).

MS. The mean square - a source's SS divided by its own degrees of freedom. Here, $MS = 2117.20$ (Between) and 49.16 (Within).

df. The degrees of freedom for a source of variance. Here, $df = 2$ (Between) and 237 (Within).

N. The number of complete cases in each group. Here, 3 groups contribute a total N of 240 (see the descriptives table for each group's own N).

Mean. The average outcome value within each group. Here, each group's mean is listed in the descriptives table above.

SD. How spread out the outcome values are within each group. Here, each group's SD is listed in the descriptives table above.

M diff. The mean difference for one pairwise comparison in the post-hoc table. Here, each pairwise contrast in the post-hoc table reports its own mean difference, using the Tukey HSD adjustment.

SE. The standard error of a pairwise mean difference. Here, each contrast's SE is listed alongside its mean difference in the post-hoc table above.

p adj. The p-value for a pairwise comparison, adjusted for multiple comparisons across all pairs. Here, pairwise comparisons use the Tukey HSD adjustment; see the post-hoc table for each pair's adjusted p.

95% CI. The range likely to contain the true mean difference for one pairwise comparison. Here, each contrast's 95% CI is shown in the post-hoc table above.

How to read this test

The F and its p tell you whether the groups differ overall; η^2 (or ω^2) is the effect size. If significant, the post-hoc table shows which specific pairs differ, with multiplicity-adjusted p-values. Your significance threshold (α) is 0.05.

APA template: A one-way ANOVA gave $F(2,237)=43.07$, $p < .001$, $\eta^2=.27$ [.16, 1.00]. Tukey HSD post-hoc tests showed...

Statistical basis: F-test for equality of several means (ANOVA) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Tukey HSD post-hoc comparison (Tukey, J. W. (1949). "Comparing individual means in the analysis of variance." Biometrics, 5(2), 99-114.); Bonferroni-corrected post-hoc comparison (a 'posthoc' choice) (Dunn, O. J. (1961). "Multiple comparisons among means." Journal of the American Statistical Association, 56(293), 52-64.); Scheffé-corrected post-hoc comparison (a 'posthoc' choice) (Scheffé, H. (1953). "A method for judging all contrasts in the analysis of variance." Biometrika, 40(1-2), 87-104.); Levene's test for equal variances (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), Contributions to Probability and Statistics (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." Journal of the American Statistical Association, 69(346), 364-367.); η^2 effect-size benchmarks (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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